

P S Mithul Sourav

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EDUCATION

National Institute of Technology Karnataka (NITK), Surathkal
B.Tech. in Mechanical Engineering; CGPA: 7.37/10

Aug 2023 – Jun 2027
Surathkal, India

RESEARCH INTERESTS

I am broadly interested in experimental and field robotics, particularly rough-terrain mobile robots, and in the dynamics, vibration, sensing, control, and autonomy of mechatronic systems.

EXPERIENCE

Vayu Aerospace Pvt. Ltd.

Jun 2026 – Jul 2026

Product Intern | UAV Vibration Analysis & Isolation

Bengaluru, India

- Compared three company-proposed flight-controller mounting architectures: a rigid baseline, an elastomer-isolated modular tray, and a suspended mount. Used ANSYS modal and response-oriented analysis to shortlist the two isolation concepts for hardware evaluation.
- Supported manufacturing, integration, and ground motor-run testing; analyzed controller IMU logs in MATLAB using comparable steady windows, RMS reduction, and FFT review, contributing to selection of the elastomer-isolated modular tray as the preferred design direction.

Thinkify Labs

May 2025 – Aug 2025

Product and Strategy Intern | Workflow Automation

Bengaluru, India

- Built an AI-based lead-qualification workflow using profile and email data to classify prospects and organize sales follow-up.
- Routed qualified prospects to appropriate sales representatives, contributing to a reported 25% improvement in operational efficiency.

ILO Consulting

May 2024 – Jul 2024

Investment Banking Intern | Financial Modeling & Market Analysis

- Built DCF, LBO, and three-statement financial models for M&A due diligence and transaction analysis.
- Contributed to an Ireland market-entry assessment covering political, regulatory, and competitive factors.

ONGOING PROJECTS

Design and Development of a Tensegrity Based Variable Stiffness Joint

May 2026 – Present

- Exploring how a tensegrity joint can change its stiffness by changing geometry and internal force distribution.
- Using MATLAB to calculate member forces, joint response, and stiffness under different geometries and external loads.
- Using these results to guide the design of a tensegrity-based joint that could later be developed into an exoskeleton or wearable robotic mechanism.

Adaptive Suspension and Vibration-Aware Control for a Rough-Terrain Rover

Sep 2026 – Present

- Designing a six-wheel rover with variable-stiffness suspension that changes spring leverage between soft, medium, and stiff settings, then locks the chosen setting mechanically.
- Using IMU and wheel-encoder data so the rover reacts to measured vibration: change stiffness when vibration stays high, check the new setting, measure again, and slow down only if needed.
- Planning the build in stages, from one suspension unit to a full rough-terrain test, then comparing fixed stiffness, speed-only control, stiffness-only control, and the combined system.

Off-Road Leaf-Collection Robot

Jul 2026 – Present

- Working from a simple but difficult problem: pick up dry or wet leaves on uneven ground without pulling in too much soil.
- Broke the robot into locomotion, pickup, transfer, storage, and terrain-following systems, then compared mechanism ideas against space, manufacturing, and integration limits.
- Built a bottom-up BOM and defined tests for traction, pickup efficiency, soil rejection, and endurance so the next design choice can be based on measured results.

COMPLETED PROJECTS

KneeAssist: Actuated Brace for Post-Operative Knee Extension Deficit

Sep 2026

- Started from the idea that the brace should help only when needed: sense knee angle, let the patient keep extending actively, and add controlled cable-and-spring assistance.
- Helped turn the CAD into a buildable system with adjustable rails and cuffs, a motor drive, angle sensing, power protection, and an independent manual release.
- Planned bench tests for angle accuracy, spring force, assisted motion, jam release, faults, and cycle life; our team was selected in the top 5 of 70 teams nationwide in the Incubate X Prosthetic Challenge.

Reaction-Wheel Microvibration Prediction with FEM and Machine Learning	Aug 2026 – Oct 2026
- Built modal and harmonic FEM models to see how reaction-wheel vibration travels through a satellite panel and reaches a camera mounting point. - Found a strong camera-response peak near 5,800 rpm, then refined the mesh to check that the result was not just a numerical artifact; the peak changed by only 0.33%. - Set up a 24-design parameter sweep and a held-out regression workflow to explore whether a faster model could still capture the structural response, especially near resonance.	
NeoLeg Passive Spring-Assisted Knee Mechanism	Dec 2025 – Mar 2026
- Explored whether a simple passive spring could assist knee motion during squatting and lifting without making the mechanism bulky or hard to move. - Built and refined the knee assembly in SolidWorks, then used ANSYS and interference checks to make sure the parts could move through the modeled flexion range. - Compared spring positions and attachment geometry to understand how small layout changes affected usable motion and the overall assist concept.	
Uncertainty-Aware Navigation for an Indoor Mobile Robot	Sep 2025 – Dec 2025
- Built a differential-drive simulator with A* path planning, heading control, wheel encoders, an IMU, and extended Kalman filter localization. - Deliberately made the simulated robot imperfect by adding encoder noise, IMU drift, yaw bias, and wheel slip, then added simple logic to slow down or recover when localization confidence dropped. - Ran Monte Carlo trials and tracked success, collisions, localization error, recovery events, and time to goal to see where the system failed, not just whether it reached the target.	
Four-Bar Linkage Door-Opening Mechanism	Feb 2024 – Apr 2024
- Built a four-bar door-opening mechanism with my team, starting from the motion we wanted and turning it into a real set of links, pivots, and joints. - Worked through link lengths and pivot locations to get coordinated door motion while staying within the kinematic limits of the mechanism. - Learned directly how a small change in link geometry can change the motion of the whole mechanism.	
Wallet for Easier Card Access	Sep 2023 – Nov 2023
- Started by noticing a simple user problem: cards in compact holders can be awkward to pull out quickly. - Developed a wallet layout around easier card access and turned that observation into a tangible mechanical design idea. - Used the project to learn a simple design rule: start with the user problem, then build the mechanism around it.	
Arduino-Based Smart Home with Solar Power	Sep 2019 – Jan 2020
- Built an Arduino-based smart home that connected several sensors and automated functions into one wired electronic system. - Designed the wiring and control logic around the sensor inputs, then debugged the connections and code until the different functions worked reliably together. - Added a photovoltaic cell power source and tested how the electronics behaved when the model was powered from solar energy instead of only an external supply.	
Smart Traffic Light System and Transit Elevated Bus Model	Sep 2018 – Jan 2019
- Built both concepts as working electronic models, wiring the circuits by hand and using logic gates to control how the systems behaved. - For the traffic-light system, created the signal sequence with logic-gate control and connected the signal lights so the junction followed a repeatable operating pattern. - For the elevated-bus model, built a separate wired control circuit for the operating sequence, then traced faults in the connections and logic until the model worked consistently.	
LEADERSHIP AND RECOGNITION	
Indian Society for Technical Education (ISTE), NITK	Sep 2024 – Present
<i>Secretary</i>	<i>NITK, Surathkal</i>
- Led sponsor outreach, budgeting, vendor coordination, and logistics for a technical event with 500+ participants. - Secured three sponsors and coordinated execution across organizational and operational workstreams.	
NH66 Fund, P&L Club	Jan 2025 – Apr 2026
<i>Fund Manager</i>	<i>NITK, Surathkal</i>
- Directed strategy for an INR 150K student-run fund and mentored three junior analysts. - Organized two recruitment drives and one training workshop for analyst onboarding and development.	
Global Case Competition at Harvard	Feb 2026 – Mar 2026
<i>Top 50 globally</i>	<i>Team of four</i>
- Competed in a four-member team and advanced to a Top 50 global placement in the 2026 competition. - Analyzed the European defence landscape and developed structured recommendations for the case.	
TECHNICAL SKILLS	
Software: MATLAB; ANSYS Mechanical; SolidWorks.	
Programming: Python; MATLAB; C; C++.	
Languages: English; Hindi; Tamil; French (basic).	